

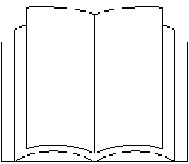


Operator's manual for Attachment



Backhoe 220 & 250

Product number: 400-700-series loaders: A21172
A21155



Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

Manufacturer:

AVANT[®]
AVANT TECNO OY
e-mail: sales@avanttecno.com

Ylötie 1
33470 Ylöjärvi
FINLAND
Tel. +358 3 347 8800
Fax +358 3 348 5511

www.avanttecno.com

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Appendix: EC Declaration of Conformity

This manual uses the following warning symbols to indicate factors that must be taken into account to reduce the risk of personal injury or damage to property:

	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or damage to property. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.
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This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This operator's manual is a translation of the original Finnish version.

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this AVANT attachment. It has been designed and manufactured on the basis of years of product development and experience. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

The use of the backhoe requires some previous experience of operating the loaders. These instructions include important instructions about mounting and using the backhoe for experienced AVANT operators as well. Ensure that all persons using the equipment have received proper guidance and familiarised themselves with the manual and all safety instructions before using the equipment. Keep this manual at hand throughout the service life of the equipment. If the manual is lost, you can request a new one from the retailer or manufacturer. If you sell or transfer the equipment, be sure to hand over this manual to the new owner.

2. Designed purpose of use

The AVANT backhoes 220 and 250 are attachments designed and manufactured to be used with AVANT multi purpose loaders. They are digging attachments with high reach, and they are suited for hard construction work, to be used in digging and loading of soil to a trailer, for example.

The slewing range of both models is full 180° with constant speed. The backhoe is easy and quick to mount on the loader, and the hydraulically operated locking bar, which mounts on the loader chassis, ensures stability during digging and enables efficient work. In addition, manually adjustable support jacks are a standard feature.

The backhoes are equipped with a hydraulic control valve unit to control the movements of the backhoe. The backhoes can be operated by standing next to the equipment, or directly from the driver's seat, when the control valve unit is mounted on the loader boom. A comprehensive range of buckets is available for various jobs. Also a grading bucket with hydraulic tilting is available, for which the backhoe is equipped with necessary components as standard. The use of the backhoe as a lifting crane is prohibited due to a risk of tipping the loader-backhoe combination over. The backhoes are not suitable for dredging.

The attachments have been designed to require as little maintenance as possible. With the backhoe, an installation kit is provided, which is mounted at the first use. Regular maintenance tasks can be performed by the operator. More demanding tasks must be left to qualified professionals. Appropriate protective equipment and original spare parts must be used for the maintenance. Read the maintenance and servicing instructions in this manual. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for the backhoes

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not familiarised yourself with its operation and the related hazards. The purpose of these instructions is to help you to:

- use the equipment in a safe and efficient manner
- monitor and prevent any hazardous situations
- keep the equipment intact and ensure a long service life



- When mounting the attachment to the loader, ensure that the locking pins of the loader's quick attach plate are in the lower position and that they **have locked the attachment to the loader. Always mount also the supporting bracket.** Carefully read the instructions of how to attach the backhoe to the loader.



- Start the use of the backhoe carefully and calmly at a safe area
- Keep others away from the operating area of the backhoe due to crushing hazard.
- Turn the auxiliary hydraulics of the loader off immediately after active use of the backhoe to prevent accidental movements of the backhoe.
- Do not use the backhoe on horizontally tilted terrain. Ensure the stability of the backhoe and the load carrying capacity of the ground even at level terrains. Adjust the outriggers correctly and use the hydraulic locking of the backhoe.
- Pay attention to the contours of the terrain and other hazards, such as rocks or branches and trees reaching the driver's seat or the backhoe control levers. Pay attention to the surroundings and any other persons and machines moving in the vicinity.
- Always transport the backhoe as low and as close to the machine as possible to keep the centre of gravity low. For transporting, turn the boom of the backhoe to its middle position and keep the bucket near the loader.
- Turn the backhoe to a position in which it will be stable during storage. See section 5.
- Shut down the loader engine before any cleaning, maintenance or adjustments. Ensure that the backhoe is supported properly supported when going near the attachment.
- Do not modify the structure of the attachment in a manner that would affect its safety.
- Also read the safety instructions for the loader in the loader manual.



Note that the backhoe and the loader boom move down even if the engine has been stopped. Never go under a raised boom or the attachment.



Familiarise yourself with the loader controls in a safe area. Pay particular attention to the safe stopping of the attachment and the machine.

Remember to wear proper protective clothing:



- The noise level in the driver's seat exceeds 85 dB(A). Wear hearing protection while working with the loader.



- The use of safety boots is recommended while working with the loader.

- Wear protective gloves when handling hydraulic hoses.

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4. Backhoe specifications

Table 1 – Backhoe 220 & 250 technical data

Product number	A21172	A21155
Digging depth:	2 200 mm	2 500 mm
Loading height:	1800 mm	2100 mm
Weight (without bucket):	350 kg	370 kg
Installation kit*:	Installation kit for 400-700 –series loaders: A48130 Installation kit for 300 –series loaders and 500-series manufactured before 2008: A48132	

*Normally the installation kit A48130 is supplied with the backhoe.

The installation for 300-series loaders and 500-series, manufactured before 2008, depends on the boom type. Contact AVANT dealer for more information.

Table 2 – Bucket options

Product number:	A2896	A2887	A2932	A2871
Width:	280 mm	400 mm	1000 mm	Grading bucket with hydraulic tilting, 700 mm

Listed below are the markings and labels fastened to the flail mower, which must be visible on the equipment. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer or contact information provided on the cover page.



The warning decals contain important safety information. Replace damaged or missing decals with new ones.

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TYPE	Axxxxx	No.	00000
Year	2010	kW	kg

- A46771 – Read the instructions before use.
- A46772 – Do not go under a raised attachment; keep a safe distance from raised or freestanding equipment.
- A46803 – Crushing hazard, keep clear of moving parts.
- A46797 – Crushing hazard in the reach area of the backhoe, keep bystanders away from danger area.

Identification plate of the attachment



A46771



A46772



A46803



A46797

5. Assembly of the backhoe

The backhoes 220 and 250 are delivered with a loader model dependent installation kit. Before the first use the installation kit must be mounted on the backhoe and the loader. When the backhoe is ready for use, mount it on the loader as shown in the next sections, and mount the bucket to the backhoe.



Beware of the hinged control valve support on the loader. Do not remove the valve unit from its stand until the backhoe is completely mounted to the loader.

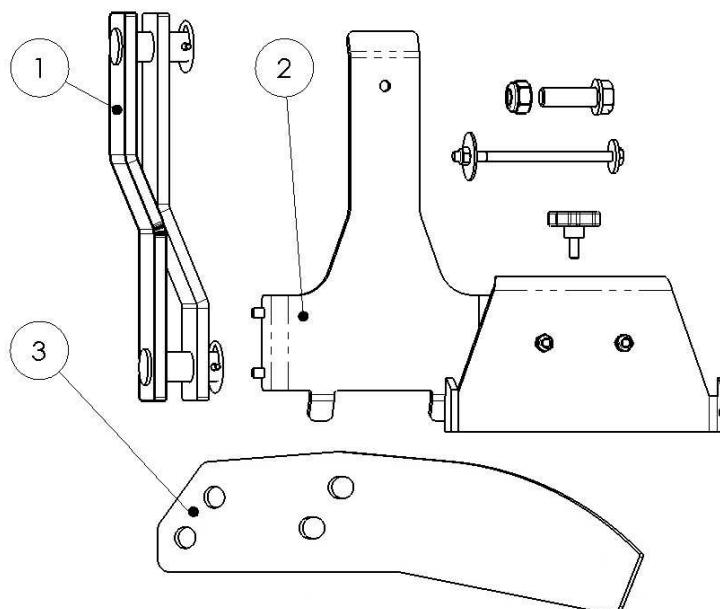
5.1. Mounting kit A48130

These instructions for the mounting kit A48130 are applicable to all 400-, 600-, and 700-series loaders, as well as to 500-series loaders manufactured in 2008 or later. Contact your AVANT dealer if using the backhoe with other loader models.

5.1.1. Contents of the mounting kit

The mounting kit contains the following loader dependent components:

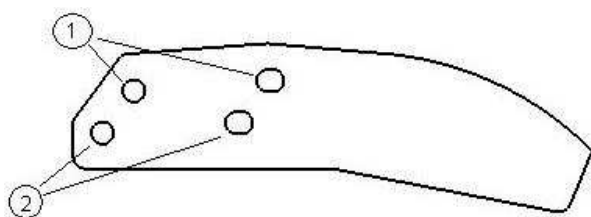
1. Supporting bracket kit to be mounted on the loader, inc. pins and locking cotter pins
2. Control valve unit mounting bracket, its adapter and mounting supplies
3. Adapters (2 pieces) for attachment hydraulic locking, bolts, washers and nuts



5.1.2. Mounting the lower adapter brackets for hydraulic locking

The backhoes are hydraulically locked to the front frame of the loader by the brackets, which are turned against the underside of the loader. Before putting the backhoe into service, the lower adapter brackets must be mounted.

The locking adapters (see figures on the right and below) are mounted to below the quick attach plate. The brackets should be mounted to the outer side of the backhoe frame brackets. The convex edge of the bracket should face upwards.



Locking adapter's mounting holes:

- 1: mounting holes for 400-600-series loaders
- 2: mounting holes for 700-series loaders

Mount the lower adapter brackets with the provided M14 sized bolts and nyloc nuts. Use washers on both the bolt and the nut side of the fixing.

5.1.3. Mounting the valve support bracket

The backhoe can be operated from two operating positions. The control valves are mounted to the loader so that it can be operated by standing next to the loader, or the valve may be mounted on the loader boom to be operated from the driver's seat. To mount the valve on the loader boom, an assembly kit must be mounted on the boom. The assembly kit can be left on the boom for easier attachment of the backhoe the next time it is used.

To make it easier to mount the assembly kit, lift the loader boom and mount the service support of the loader boom. Take up the items shown in the adjacent figure.

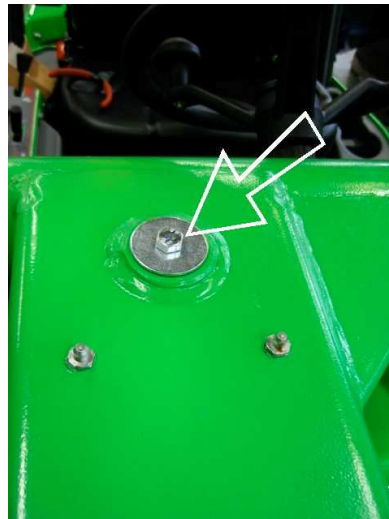


Use the service support of the loader boom during installing of the valve mounting bracket.



Remove the pin holding screw from the upper end of the loader boom, shown in the left figure.

Mount the holding bracket on the boom and replace the removed screw with the longer screw provided with the kit.



When the holding bracket is in its place, tighten the screws on the right side of the bracket against the loader boom.

Do not overtighten the screws!

Fit the valve holder to the holding bracket as shown in figure below.



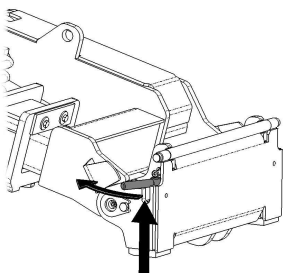
5.2. Mounting the backhoe to the loader

Mounting the backhoe to the loader must be done carefully. The backhoe may tip over on the loader or the operator, if it has not been properly attached. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions in Section 3.

The attachment is mounted to the loader boom by using the quick attach plate and the counterpart on the attachment. In addition to this, a locking bracket must be also installed. The backhoe is mounted to the loader as follows:

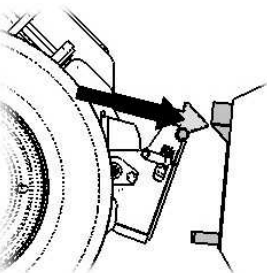


Never move or raise an attachment that has not been locked.



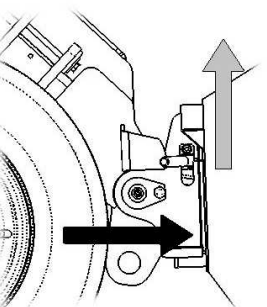
Step 1:

- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- Ensure that the hydraulic hoses are not in the way during installation.



Step 2:

- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this.



Step 3:

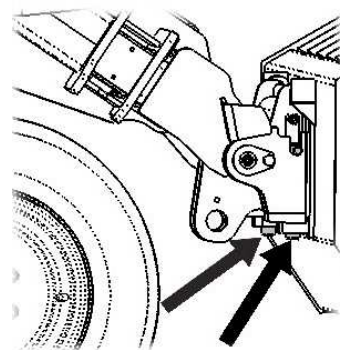
- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Lock the locking pins manually – do not use the automatic returning of the locking pins.

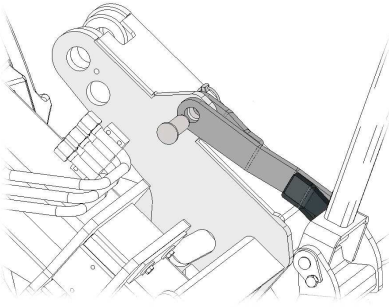


Do not use the automatic locking of locking pins with the backhoe. The centre of gravity of the backhoe is at relatively high, which creates a risk of the backhoe tipping over. Lower the attachment on the ground and secure the locking manually.



Before moving or lifting the attachment, ensure that the locking pins are in the lower position and come through the fasteners on both sides.





Step 4: Fitting the support bracket

- Lift the backhoe slightly off the ground. Remove the pin from the loader side of the supporting bracket and place the bracket on the bracket of the loader boom.
- Tilt the backhoe to align the holes.
- Fit the locking pin through the holes and lock it with the cotter pin.



Do not use your fingers to check the alignment of the support bracket. Movement of the backhoe creates a pinch or shear hazard.

5.2.1. Taking the backhoe off from the loader

When taking the backhoe off from the loader for storage, to ensure its stability, do as follows:

1. Open the hydraulic locking completely (see section 5.4)
2. Lower the outriggers of the backhoe on even and solid surface.
3. Turn the slewing of the backhoe to its middle position and lower the bucket just slightly above the ground, as shown in the adjacent figure.
4. Turn the auxiliary hydraulics control of the loader off.
5. Put the control valve of the backhoe to its holder on the backhoe. Note that the backhoe may move if the control levers are actuated.
6. Turn the loader boom tilting so that it is possible to remove the pin of the support bracket.
7. Shut down the loader engine and remove the support bracket
8. Tilt the backhoe to a slightly forward position so, that the bucket will now lower on the ground.
9. Disconnect the hydraulic fittings as shown in section 5.3. This procedure will release the pressure left in the backhoe and the bucket and the backhoe booms will lower to a stable position.
10. Open the quick coupling and take the backhoe off from the loader.



Backhoe storage position

5.3. Connecting and disconnecting the hydraulic fittings

Before connecting or disconnecting hydraulic fittings, shut down the loader engine and move the auxiliary hydraulics lever on the loader to its extreme positions a few times and also move the control levers of the backhoe. This way, there will be no remaining pressure in the system and the fittings are easy to connect.

To connect and disconnect the fittings, move the collar at the end of the female fitting. The hoses should be connected so that the male fitting of the backhoe is connected on the loader to the fitting with coloured cap. The second hose is connected on the single male fitting of the loader.

When disconnecting the fittings, first move the auxiliary hydraulics control levers as well as the levers of the backhoe. Lower the backhoe as described in the previous section so that there will be no residual pressure in the system and the backhoe will be stable during storage.

If there is residual pressure in the hydraulic system of the backhoe due to its position, it is often possible to disconnect the fittings, but connecting them the next time may be difficult. If the fittings will not connect, the residual pressure must be released. First move the control levers of the backhoe. Note that the backhoe may move. If this doesn't help, while the backhoe is coupled with the loader, its bucket may be pressed against the ground with the loader boom.

In other cases the pressure left in the system must be relieved by opening the hose fitting slightly with tools. Keep both of the hoses disconnected and wrap the fitting in a cloth to prevent oil spillage. When the fitting is opened slightly, a small amount of oil will be released; beware of the oil. Wear protective gloves and goggles. Tighten the fitting immediately after the procedure.



The backhoe may move when the pressure is released, stay clear from the danger area. To prevent oil spillage wrap the fitting with cloth when releasing pressure from the hoses.



When removing the attachment, always disconnect the hydraulic fittings before detaching the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.



Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure. High-pressure fluid ejection may cause injuries.

Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. Never leave the hoses hanging on the ground; place them on top of the equipment.

5.4. Control valve of the backhoe

In larger excavation tasks and constant changing of location it is advisable to mount the control valve on the bracket on the loader boom that is included in the installation kit of the backhoe. The backhoe may be operated from the control position next to it in tasks demanding accuracy or when visibility is hindered, or when using the backhoe only for a little while. Move the control valve back to its place on the backhoe before taking the backhoe off from the loader.



To prevent unintended movements of the backhoe shut down the loader and lower the bucket of the backhoe onto ground before starting to move the valve. Make sure that the valve is securely locked on the selected operating position with the locking screw.



On loaders equipped with cabin L, LX or DLX, the lower section of the windscreen must be removed to make it possible to mount the valve on the boom.

5.5. Mounting a bucket on the backhoe

The buckets are connected to the backhoe with two pivot pins. Change the bucket when the backhoe is mounted to the loader. Place the bucket near the ground. Ensure that the backhoe is adequately supported. Hold the bucket while removing it and use a soft hammer or a block of wood to soften the hits, so that the pins will not get damaged. Do not hit directly at the pins.

5.6. Grading bucket

With the grading bucket it is easy to level the ground surface to desired slopes or to dig a ditch. The backhoes are equipped with necessary components to fit and operate the grapple bucket; the control valve has the needed block and lever for the bucket. To mount the grading bucket, also hydraulic quick couplings should be mounted on the backhoe. Contact service if needed.

6. Instructions for use

When the installing kit of the backhoe and the backhoe are mounted on the loader the work can be started. The control levers of the backhoe will operate once the auxiliary hydraulics of the loader has been locked on. Use the outriggers and the hydraulic locking of the backhoe; their operation is shown on the following pages.

Check the operating area and the attachment once more before starting work.

- Check that the backhoe is intact and that all obstacles have been removed from the working area before operation. Remember correct working methods and avoid unnecessary descending from the loader.
- Make sure that it safe to dig in the location. Find out if there are electric cables, water lines or similar obstacles at a depth that can be reached with the digger.
- Operate the backhoe only at well-lit areas.
- Ensure that the loader and attachment are used in a safe manner and as intended.
- Check the hydraulic hoses and their connections and that the hoses are not subjected to stretching or squeezing at any position of the backhoe. Check regularly to ensure there are no hydraulic leaks. Refer to Section 7 for maintenance instructions.
- Operate the attachment only from the driver's seat or the operating position at next to the backhoe. Do not let others to use the controls when working near the backhoe.



Turn the auxiliary hydraulics off whenever not operating the backhoe or when transporting the backhoe. The control levers of the backhoe may be accidentally operated, so that accidental movement of the backhoe may create a hazardous situation.



Never go under a raised boom or the attachment. The backhoe components and the loader boom move down even if the engine has been stopped.

6.1. Using the outriggers

The backhoe is equipped with outriggers, which are intended to stabilize the backhoe during digging and moving material to the sides. Using the backhoe is more efficient with the use of the outriggers, so it is advisable to set them correctly. The outriggers are mounted so that its foot points forward from the loader. Note that the left and the right ones are dissimilar.

The correct setting of the outrigger will also affect the hydraulic locking shown in the next section. The height of the outriggers should be adjusted according to the terrain. There are several holes on the backhoe frame and the outrigger. Use the cotter pin to lock the adjustment pin.

6.2. Using the hydraulic locking of the backhoe

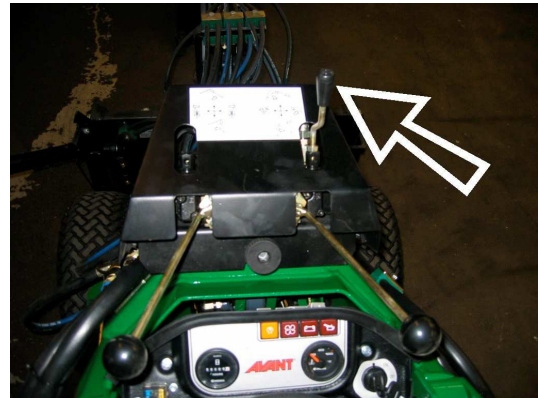
For more rigid mounting of the backhoe, it is locked onto the loader front frame by hydraulically operated brackets. When the backhoe, its support bracket and hydraulic hoses are connected, the **telescopic boom of the loader must be retracted completely**.

Lower the loader boom to as down as possible, which will lead to raising the loader front wheels off the ground about 10 cm, if the outriggers are adjusted correctly. When using the hydraulic locking, the weight of the front of the loader will be on the outriggers, giving more stability. Note that because of the support bracket, on loaders equipped with the self-parallel system, the loader boom control lever should also be moved sideways to lower or to lift the backhoe.

The locking is operated with the extra lever on the right of the control valve, shown in figure on the next page. The lever will move the brackets mounted under the backhoe and they are intended to be turned against the bottom of the loader front frame.



Before the first use ensure that the locking brackets are adjusted so that they lean against even surface under the loader front frame, and not against bolt ends or similar edges. Adjust the brackets if needed.



For the best stability use the outriggers and the locking as described.



To avoid necessary load being applied on the loader boom structures do not try to operate the loader boom while the locking is engaged. Open the hydraulic locking before using the boom control lever.

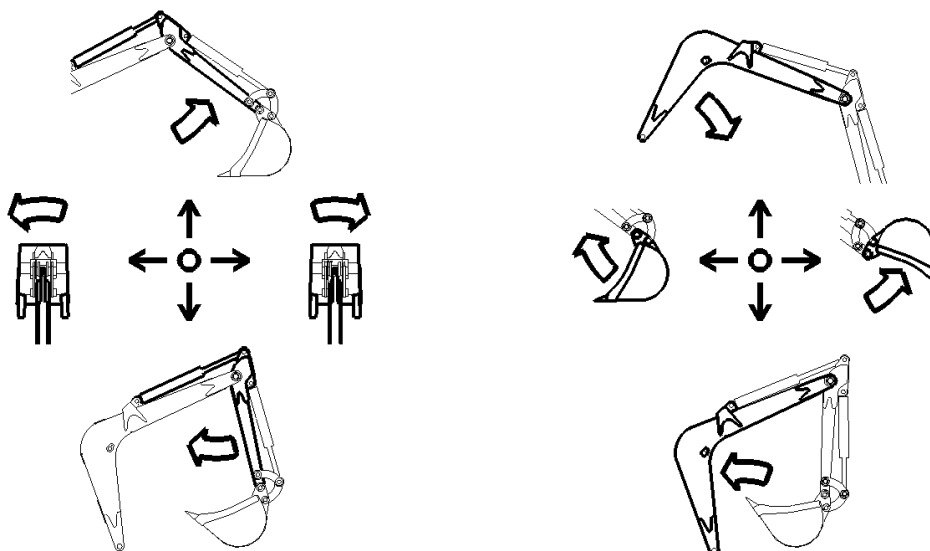
6.3. Digging with the backhoe

Check the operating area before starting to work:



- Ensure that there are no other persons at the work site. Do not allow others closer than five meters from the backhoe, near the danger area of the boom or directly in front of the backhoe.
- Make sure that it is safe to dig at the site. Find out if there are any electric cables, water lines or similar obstacles within the reach of the equipment.

The backhoe is controlled with its two control levers. The operation of the control levers is shown in the figure below. The backhoe has relatively many simultaneous movements, and it is advisable to familiarise the operation with time and away from other persons or buildings.



Backhoe controls, left and right control lever



Beware of the caving in hazard at the excavation and around it. Do not drive near the edge of the hole or allow children to play near it.



Do not operate the backhoe at horizontally overly tilted terrains. Transport the backhoe as low and close to the loader as possible. Drive carefully on inclined terrains.

Drive the loader so that digging is performed from front of the loader. On inclined terrains avoid excess pulling from the sides, which can affect the stability of the loader.

Digging with the backhoe is the most efficient when the edge of the bucket is kept at an angle that allows the edge to cut the soil the most effectively. Pull the edge of the bucket towards the loader with the left lever and tilt the bucket by operating the right control lever at the same time. The bucket should not be pressed too hard downwards, since this would result in lifting the front tires of the loader, and digging becomes ineffective.

When operating with the backhoe, moderate engine revolutions are often enough. Higher revolutions will not increase the digging force. Keeping the engine revolutions moderately low will save fuel; keep the revolutions just high enough so that the engine will not stall.

At the first use the movements of the backhoe may be imprecise as there might be air trapped in the hydraulic components. The hydraulic system will bleed itself, when moving all the movements to their extreme positions.



When lifting a load from a high level, the loader could tip forward when reversing with the loader. Do not reverse the loader while simultaneously dragging the load. Make sure that the loader can lift and handle the load being lifted. Drive carefully on rough terrains and keep the load as low and as close to the loader as possible.



Do not use the backhoe without the supporting bracket. The purpose of the bracket is to lock the tilting and to make the backhoe more stable for more effective operation.

7. Inspections, maintenance and servicing

Continuous maintenance of the backhoe includes regular cleaning and lubricating. The condition of the hydraulic hoses and components should be monitored. Due to crushing hazard caused by the moving components of the backhoe, all maintenance work must be done when the backhoe has been lowered down completely, its residual pressure is released and the loader is shut down.



Ensure that the backhoe is properly supported before any maintenance work. Note that the backhoe may move also when the loader has been shut down and watch out pinching of fingers or hands in particular.

Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Do not use the equipment, if you have discovered a fault in the hydraulic system. Leaking hydraulic fluid may penetrate the skin and cause serious injuries. Hydraulic fluid is also harmful to the environment.

Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of the hose has worn off. If there are signs of leaking, to check the component, hold up a piece of cardboard in the area where a leak is suspected. Finding any of these faults means that the hydraulic hose or component in question must be replaced. Spare parts are available from your nearest AVANT retailer or service point.



Never tighten a hydraulic fitting under pressure, since the fitting may break and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.



7.1. Cleaning the equipment

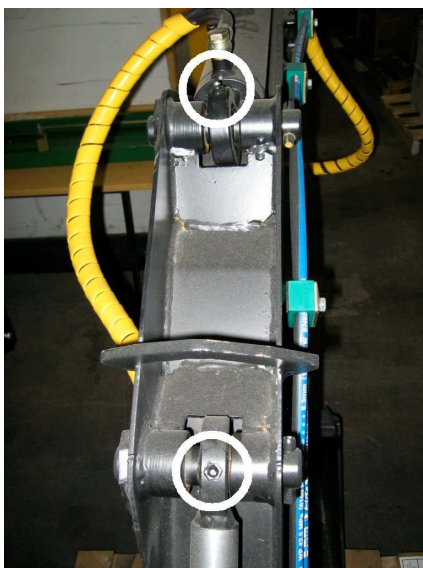
You may use a pressure washer and a mild detergent to clean the backhoe. Do not use strong solvents or spray directly at the hydraulic components or at the labels on the attachment either. Lubricate the lubrication points on the attachment after washing.

Do not let water to stay in the bucket. For longer storage periods touch up the paint where needed, grease the lubrication points and oil the visible parts of hydraulic cylinder shafts to prevent rust damage. Do not store the attachment directly against the ground; place it on blocks of wood or on a pallet, for example.

7.2. Lubricating

The lubricating of the backhoe consists of lubricating the 14 lubricating points of the backhoe and the slewing chain. A small amount of lubricant should be added every 5-10 hours of use. The locations of the lubrication points are shown in the figures below. All lubrication nipples are standard R1/8" nipples, replace any damaged nipples.

Lubricate the also the slewing chain with suitable oil when the lubricating points are greased. Lubricate the chain also when storing the backhoe for longer storage periods to prevent rust damage.



Lubricating points of the upper part of the main boom



Lubricating points of the lower part of the main boom



Lubricating points of the slewing joint



Lubricating points of the bucket linkage



Lubricating points of the front boom



Lubricant must be added approximately every 5-10 operating hours. Clean the end of the nipple before greasing. Only add the grease amount of a couple of grease press pushes at a time.

8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) for the attachment from the date of purchase.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work, such as the replacement of blades or the belts, or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus
EG-försäkran om överensstämmelse
EC Declaration of Conformity
EG-Konformitätserklärung



Valmistaja / Tillverkare / Manufacturer / Hersteller:
Osoite / Adress / Address / Adresse:

AVANT TECNO OY
 Ylötie 1
 33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:
 SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 474-4

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats:
 SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 474-4

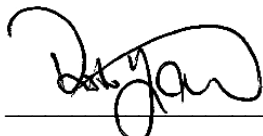
We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:
 SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 474-4

Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen). Die folgende harmonisierte Normen wurden angewandt:
 SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 474-4

Kuvaus: Avant-kuormainten kanssa käytettävä kaivuulaite
 Beskrivning: Grävaggregat; arbetsredskap för Avant lastare
 Description: Backhoe bucket; attachment for Avant loaders
 Beschreibung: Anbaubagger; Anbaugerät für Avant Radlader

Mallit / Modeller / Models / Modellen:

Avant		
Kaivuri Grävaggregat Backhoe Anbaubagger		
Mallit / Märken / Models / Modellen:		300-700 series
220	ilman kauhaa utan skopa without bucket ohne Schaufel	A21172
250	ilman kauhaa utan skopa without bucket ohne Schaufel	A21155

Allekirjoitus / Namnteckning / Signature / Unterschrift:  10.6.2010 Ylöjärvi, Finland

Risto Käkelä,
 Toimitusjohtaja / Verkställande direktör /
 Managing Director / Geschäftsführer





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